

Method 4: INTEGRATING FACTOR METHOD

Lesson 22

If a differential eqn can be expressed in the form:

$$\frac{dy}{dx} + p(x)y = f(x) \quad \leftarrow \text{Linear form}$$

then we can apply I.F.-method:

Example 1

Solve $\frac{dy}{dx} - y = x$

Solution

$\frac{dy}{dx} - y = x$, this is linear form

Step 1: $p(x) = -1$, $\int p(x) dx = \int -1 dx = -x$

IF = $e^{\int p(x) dx} = e^{-x}$

Step 2: Multiply by IF both sides of Linear d.e:

$$e^{-x} \left(\frac{dy}{dx} - y \right) = x \cdot e^{-x}$$

Step 3: $\frac{d}{dx} (y \cdot e^{-x}) = x e^{-x}$

Integration by parts

$u = x$ $v = e^{-x}$

$u' = 1$ $v' = -e^{-x}$

Step 4: Integrate both sides: $y \cdot e^{-x} = \int \underbrace{x}_{u} \underbrace{e^{-x}}_{v'} dx + C$

$$y \cdot e^{-x} = -x e^{-x} - e^{-x} + C$$

$$y = -x - 1 + C e^x$$

$$y = C e^x - x - 1$$

Example 2

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Solve $\frac{dy}{dx} + 3y = \sin x$

Solution

Here $p(x) = 3$; $\int p(x) dx = \int 3 dx = 3x$; $I.F = e^{3x}$

$$e^{3x} \left(\frac{dy}{dx} + 3y \right) = e^{3x} \sin x$$

$$\begin{cases} u = e^{3x} & v = \cos x \\ u' = 3e^{3x} & v = \sin x \end{cases}$$

$$\frac{d}{dx} (y \cdot e^{3x}) = e^{3x} \sin x$$

$$y e^{3x} = \int e^{3x} \sin x + C$$

Integration by parts

$$u = e^{3x}, \quad v = \sin x$$

$$u' = 3e^{3x}, \quad v = -\cos x$$

$$e^{3x} y = \frac{1}{10} e^{3x} (3 \sin x - \cos x) + C$$

$$I = -e^{3x} \cos x + 3 \int e^{3x} \cos x dx$$

$$I = -e^{3x} \cos x + 3 e^{3x} \sin x - 9I$$

$$10I = e^{3x} (3 \sin x - \cos x)$$

$$I = \frac{1}{10} e^{3x} (3 \sin x - \cos x) + C$$

$$\therefore y = \frac{1}{10} (3 \sin x - \cos x) + C e^{-3x}$$

Example 3

Solve $x \frac{dy}{dx} + y = x^3$

Solution

(Step 1) Make it linear: $\frac{dy}{dx} + \frac{1}{x} y = x^2$

$$p(x) = \frac{1}{x} ; \int p(x) dx = \int \frac{1}{x} dx = \ln(x) ; I.F = e^{\ln(x)} = x$$

$$x \left(\frac{dy}{dx} + \frac{1}{x} y \right) = x^3$$

$$\frac{d}{dx} (y \cdot x) = x^3$$

$$y \cdot x = \int x^3 dx = \frac{1}{4} x^4 + C$$

$$y = \frac{1}{4} x^3 + \frac{C}{x}$$

Example 4

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$$\text{Solve } (x+1) \frac{dy}{dx} + y = (x+1)^2$$

Solution

Make it linear $\frac{dy}{dx} + \frac{1}{x+1} y = x+1$ ← Linear

$$P(x) = \frac{1}{x+1} ; \int P(x) dx = \int \frac{1}{x+1} dx = \ln(x+1) ; IF = e^{\ln(x+1)} = x+1$$

$$(x+1) \left[\frac{dy}{dx} + \frac{1}{x+1} y \right] = (x+1)^2$$

$$\frac{d}{dx} (y \cdot (x+1)) = (x+1)^2$$

$$y \cdot (x+1) = \int (x+1)^2 dx$$

$$y \cdot (x+1) = \frac{1}{3} (x+1)^3 + C$$

$$y = \frac{1}{3} (x+1)^2 + \frac{C}{x+1}$$

Example 5

$$\text{Solve } (1-x^2) \frac{dy}{dx} - 2xy = 1$$

Solution

$$\frac{dy}{dx} - \frac{2x}{1-x^2} y = \frac{1}{1-x^2} ; P(x) = \frac{-2x}{1-x^2} ; \int P(x) dx = \int \frac{-2x}{1-x^2} dx = \ln(1-x^2)$$

$$IF = e^{\ln(1-x^2)} = 1-x^2$$

$$(1-x^2) \cdot \left[\frac{dy}{dx} - \frac{2x}{1-x^2} y \right] = 1$$

$$\frac{d}{dx} (y \cdot (1-x^2)) = 1$$

$$y \cdot (1-x^2) = \int 1 dx = x + C$$

$$y = \frac{x}{1-x^2} + \frac{C}{1-x^2}$$

Example 6

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$$\text{Solve } x \frac{dy}{dx} - 5y = x^7$$

Solution

$$\frac{dy}{dx} - \frac{5}{x}y = x^6 \quad \text{linear}, \quad p(x) = -\frac{5}{x}; \quad \int p(x) dx = \int -\frac{5}{x} dx = -5 \ln x = \ln(x^{-5}) = \ln\left(\frac{1}{x^5}\right)$$

$$\frac{1}{x^5} \left(\frac{dy}{dx} - \frac{5}{x}y \right) = x^6 \cdot \frac{1}{x^5} \quad \text{IF} = e^{\ln\left(\frac{1}{x^5}\right)} = \frac{1}{x^5}$$

$$\frac{d}{dx} \left(y \cdot \frac{1}{x^5} \right) = x$$

$$y \cdot \frac{1}{x^5} = \int x dx = \frac{1}{2} x^2 + C$$

$$\underline{\underline{y = \frac{1}{2} x^7 + C x^5}}$$

Example 7

$$\text{Solve: } (1-x^2) \frac{dy}{dx} - xy = 1$$

Solution

$$\frac{dy}{dx} - \frac{x}{1-x^2} y = \frac{1}{1-x^2}; \quad p(x) = -\frac{x}{1-x^2}; \quad \int p(x) dx = \frac{1}{2} \int \frac{-2x}{1-x^2} dx = \frac{1}{2} \ln(1-x^2) = \ln(\sqrt{1-x^2})$$

$$\sqrt{1-x^2} \left[\frac{dy}{dx} - \frac{x}{1-x^2} y \right] = \frac{\sqrt{1-x^2}}{1-x^2}$$

$$\text{IF} = e^{\ln(\sqrt{1-x^2})} = \underline{\underline{\sqrt{1-x^2}}}$$

$$\frac{d}{dx} \left(y \cdot \sqrt{1-x^2} \right) = \frac{1}{\sqrt{1-x^2}}$$

$$y \cdot \sqrt{1-x^2} = \int \frac{dx}{\sqrt{1-x^2}} = \sin^{-1} x + C$$

$$\underline{\underline{y = \frac{1}{\sqrt{1-x^2}} \sin^{-1} x + \frac{C}{\sqrt{1-x^2}}}}$$